ABDALLAH YEHYA SOLIMAN

AI Engineer | Data Scientist

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PROFESSIONAL SUMMARY

Computer Science graduate and AI Engineer building LLM applications, agentic workflows and data-driven products. Project experience includes natural-language-to-SQL, automated job discovery, wellness application prototypes and computer vision. Combines Python and SQL with full-stack development, model evaluation, automated testing and failure handling to build practical AI workflows.

PROFESSIONAL EXPERIENCE

Web Developer (Freelance) | Modern Broadcast Center

Jan 2025 - Mar 2025

Website: mbcegypt.com

- Developed and deployed a full-stack web application using Next.js, Prisma ORM and PostgreSQL through Supabase, implementing product management and checkout workflows.
- Implemented responsive layouts and optimized application performance across devices.

Backend Intern | Trans-IT

Sep 2023 - Nov 2023

- Designed and managed relational databases using SQL Server, developing a Supermarket Management System covering inventory and sales operations.
- Applied data structures, algorithms and object-oriented programming to backend development.

Business Intelligence Intern | Coca-Cola Egypt

Aug 2021 - Sep 2021

- Applied machine learning techniques to an AI analytics project supporting fraud detection.
- Built and trained an image classification model to recognize product types, including data preprocessing, feature extraction and model evaluation.
- Analyzed structured datasets and created interactive Power BI dashboards to support business decisions.

EDUCATION

Bachelor’s Degree in Computer Science

Ain Shams University | 2020 - 2024 | Department: Computer Science | GPA: 2.95

GRADUATION PROJECT

Smart School System

Graduation project | Ain Shams University | Grade: A

- Designed and implemented an end-to-end AI system applying machine learning and deep learning to student identification, attendance and safety monitoring.
- Developed a face recognition pipeline using a Siamese network to identify students.
- Integrated a liveness detection model into the attendance workflow to detect spoofing attempts and improve system reliability.
- Built a violence detection model using sequence-based deep learning to monitor behavior and trigger alerts.
- Implemented a uniform-compliance classification model using supervised learning to identify whether students meet school uniform requirements.

SELECTED PROJECTS

Agentic NL2SQL Analytics Platform

Next.js | TypeScript | PostgreSQL | Supabase | LLMs

Live demo: nl-2-sql-seven.vercel.app | GitHub repository

- Built an agentic system that converts natural-language questions into executable SQL across a multi-table relational database.
- Implemented metadata-driven schema routing and sub-schema selection to identify relevant tables, columns and relationships before SQL generation.
- Designed SQL generation, validation, self-correction and natural-language answer workflows for database querying.

JobHunt

Personal agentic job-discovery and matching system | Python | SQLite | GitHub

- Built a local agent scheduled to run daily at 8 AM, using CV, skills and project context to discover and rank relevant job opportunities.
- Combined career-site sources and read-only job-alert email ingestion with normalization, deduplication, eligibility filters and daily reports.
- Added deterministic scoring with LLM review, Gemini-to-Groq fallback and source timeouts to keep individual failures from blocking the pipeline.

AI Wellness Egypt

Wellness application prototype | FastAPI | Next.js | Supabase | Docker | GitHub

- Implemented consent-based onboarding, mood check-ins and a guarded chat workflow with feature flags, rate limiting and source citations.
- Added fail-closed safety checks and privacy-conscious logging; verified chat flows locally using mock sign-in, a mock provider and synthetic content. Clinical and production validation remain pending.

Fair Price Egypt

Price intelligence prototype | Synthetic and mock data | GitHub

- Built Egyptian product search, product matching and VAT-aware price comparison using synthetic demonstration data, with provider approval gates for future ingestion.
- Implemented an Arabic RTL photo-search flow with upload validation and candidate confirmation; identification uses a mock for bundled sample images.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript, JavaScript, Java, C++, C#

AI and ML: Machine learning, deep learning, NLP, computer vision, feature engineering, model evaluation; Scikit-learn, TensorFlow, PyTorch, NLTK

LLM applications: Prompt engineering, AI agents, NL2SQL, schema routing, SQL validation, self-correction, LLM provider fallback

Data and development: PostgreSQL, Supabase, SQL Server, SQLite, Next.js, FastAPI, Prisma ORM, Docker, Git/GitHub, Power BI

Conceptual knowledge: Retrieval-augmented generation (RAG), embeddings, semantic search

LANGUAGES

Arabic: Native | English: Professional proficiency